

Abeer Miglani

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EDUCATION

Shiv Nadar Institution of Eminence

Bachelor of Technology in Electrical and Computer Engineering

Dadri, UP

Aug. 2025 – May 2029

EXPERIENCE

Technical Project Trainee – Special Projects

LOHUM Cleantech Pvt. Ltd.

Jun. 2023

Kasna, UP

- Built an automated metal-price scraper in Python using Selenium WebDriver to collect pricing across multiple commodity pages.
- Used XPath selectors and explicit WebDriver waits to reliably extract content from dynamically rendered pages.
- Cleaned, structured and exported pricing data with Pandas to CSV for downstream analysis.
- Generated a lightweight HTML dashboard to visualize and share the collected pricing data.

PROJECTS

Redis-Compatible Server | C++, POSIX Sockets [GitHub](#)

Ongoing

- Building a Redis-compatible in-memory key-value server in C++ from first principles, following *Build Your Own Redis with C/C++*.
- Implemented TCP client-server communication directly on POSIX sockets.
- Designed a length-prefixed binary protocol (4-byte header + payload) to frame messages over a TCP byte stream, with read/write loops that handle partial reads and writes and a 4 KB message-size guard.
- Next on the roadmap: event loop with non-blocking I/O, hashtable-backed key-value store, TTL expiry and a thread pool.

Ripple – Cascading Infrastructure Failure Simulator | Python, FastAPI, Neo4j, React, Rust [GitHub](#) Sep. 2026

- Sole developer for team SatishSystemsInc at Manipal Hackathon 2026 (Disaster Resilience track).
- Built a Motter-Lai overload cascade engine on an in-memory NetworkX graph of power, water, transit and telecom assets, streaming each failure wave to a React/deck.gl map over WebSockets.
- Designed a recommendation engine that verifies every intervention (capacity hardening, redundancy links) by re-running the cascade, ranked by failures prevented, hospitals kept online and population spared.
- Wrote a Rust (PyO3) extension for the all-pairs shortest-path hotspot, called 20+ times per recommendation request; ran PostgreSQL/PostGIS, Neo4j, Redis and Celery with Docker Compose and GitHub Actions CI.

Terminal-Based Quiz System | Python, MySQL, SQL

Jan. 2025 – Feb. 2025

- Built a terminal-based quiz system in Python with MySQL-backed persistence.
- Designed SQL schemas and queries to store quiz questions and user responses.
- Implemented backend logic for quiz flow, scoring and data retrieval.

PROBLEM SOLVING

- Solved 100+ data structures and algorithms problems across LeetCode, Codeforces, CodeChef and HackerRank, including 95 on LeetCode.
- Continuing independent practice in data structures, algorithms and C++ problem solving.

TECHNICAL SKILLS

Languages: C++, Python, C, SQL, TypeScript, HTML/CSS; learning Rust

Frameworks & Libraries: FastAPI, Flask, React, Celery, NetworkX, Pandas, NumPy, Selenium

Databases: PostgreSQL/PostGIS, Neo4j, Redis, MySQL

Developer Tools: Git, Docker, GitHub Actions, Google Cloud Platform, VS Code

Areas of Interest: Systems Programming, Computer Networks, Data Structures and Algorithms